Permutations and Combinations

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the number of ways to arrange the 7 letters in the word SUCCESS.

2 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 5 people be seated in a row if two particular people (A and B) must not sit next to each other?

3 [Foundational] • Target: Permutations & Combinations Practice Exercises

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find the coordinates of the centre and the exact radius of C. (b) Find the equation of the tangent to C at the point $P(6, 2)$.

4 [Foundational] • Target: Permutations & Combinations Practice Exercises

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

5 [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

6 [Foundational] • Target: Permutations & Combinations Practice Exercises

Solve the equation $2\cos 2x = 1 + \sin x$ for $0 \le x \le 2\pi$ radians, giving answers to 3 significant figures.

7 [Foundational] • Target: Permutations & Combinations Practice Exercises

A block of mass $8\ \text{kg}$ rests in limiting equilibrium on a rough plane inclined at 30° to the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) calculate the normal reaction R, (b) find the coefficient of friction μ between the block and the plane.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Permutations & Combinations Practice Exercises

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

9 [Intermediate] • Target: Permutations & Combinations Practice Exercises

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).

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10 [Intermediate] • Target: Permutations & Combinations Practice Exercises

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

11 [Intermediate] • Target: Permutations & Combinations Practice Exercises

A train accelerates uniformly from rest to 30 m/s in 20 s, travels at constant speed 30 m/s for 60 s, and then decelerates uniformly to rest in 15 s. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

12 [Intermediate] • Target: Permutations & Combinations Practice Exercises

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

13 [Intermediate] • Target: Permutations & Combinations Practice Exercises

A block of mass 4 kg is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude 25 N acting at 30° above the horizontal. Taking $g = 9.8$ m/s^2: (a) show that the normal reaction $R = 26.7$ N, (b) find the acceleration of the block.

14 [Intermediate] • Target: Permutations & Combinations Practice Exercises

Two particles P of mass 3 kg and Q of mass 5 kg are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8$ m/s^2: (a) find the acceleration a of the system, (b) calculate the tension T in the string, (c) find the force exerted by the string on the pulley.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

A continuous random variable X has PDF given by $f(x) = \frac{k}{x^2}$ for $1 \le x \le 4$, and 0 elsewhere. (a) Show that $k = \frac{4}{3}$. (b) Find the median of X.

16 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

17 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

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18 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

19 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.

20 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Find the equation of the perpendicular bisector of the line segment joining $P(2, -1)$ and $Q(8, 7)$. Give your answer in the form $y = mx + c$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{5}P_{2} \).

a. 25

b. 21

c. 10

d. 20

22 [Foundational] • Target: Permutations & Combinations Practice Exercises

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^nC_r = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?

a. 10

b. 15

c. 20

d. 60

23 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many distinct ways can 6 different people be seated around a circular table (rotations considered identical)?

a. 120

b. 103

c. 122

d. 100

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24 [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the number of distinct arrangements of the letters of the word PUPPY.

a. 103

b. 27

c. 64

d. 60

25 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 8 different books be arranged in a row on a shelf?

a. 45808

b. 35108

c. 40625

d. 40320

26 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 6 different people be seated in a row of 6 chairs?

a. 720

b. 817

c. 800

d. 725

27 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 11 different people be arranged in a row if two particular people must NOT sit next to each other?

a. 32806546

b. 33364823

c. 34615086

d. 32659200

28 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 8 different people be arranged in a row if two particular people must sit together?

a. 10080

b. 8546

c. 9551

d. 9288

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Permutations & Combinations Practice Exercises

In how many ways can 6 different runners be lined up at the start of a race if two particular runners must stand next to each other?

a. 273

b. 240

c. 220

d. 241

30 [Intermediate] • Target: Permutations & Combinations Practice Exercises

In how many ways can 5 of 7 different people be selected and arranged in a row?

a. 2891

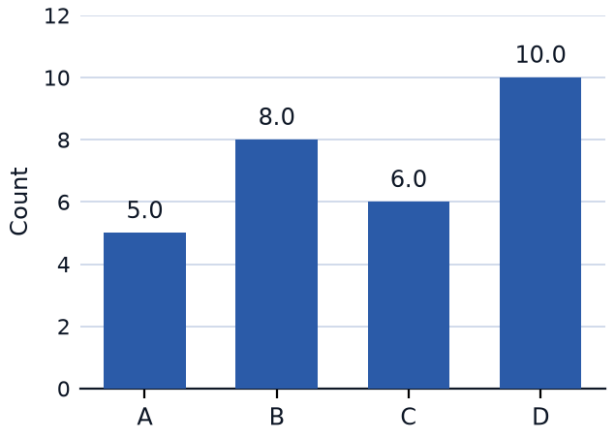
b. 2282

c. 2520

d. 2348

31 [Intermediate] • Target: Permutations & Combinations Practice Exercises

The histogram below shows the distribution of heights of a group of students. The total number of students in the group is 50. If the frequency density for the class interval $60 \leq h < 70$ is 1.2, find the value of k.



a. 10

b. 12

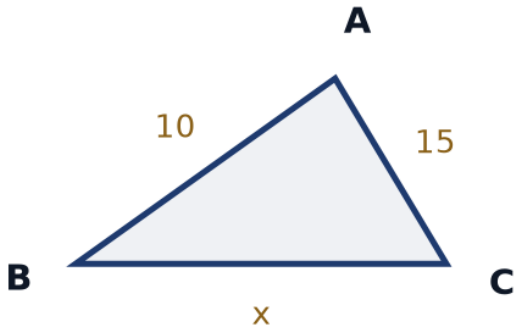
c. 15

d. 20

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32 **[Intermediate]** • **Target: Permutations & Combinations Practice Exercises**

In triangle ABC, angle ABC = θ degrees, and angle ACB = 2θ degrees. The lengths of sides AB and AC are 10 cm and 15 cm respectively. The length of BC is denoted by x cm. Given that the area of triangle ABC is 60 cm^2, find the value of x, to 3 significant figures.



a. 13.7

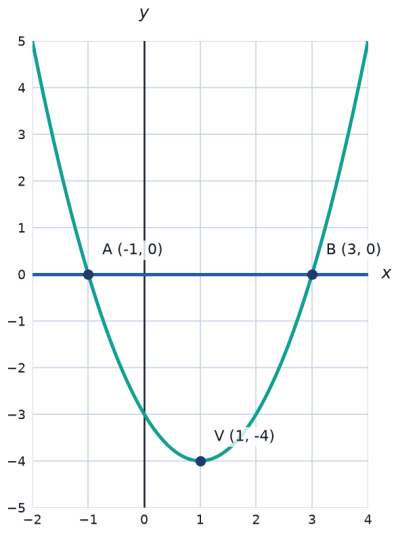
b. 11.4

c. 12.8

d. 14.9

33 **[Intermediate]** • **Target: Permutations & Combinations Practice Exercises**

The diagram shows a curve with equation $y = ax^2 + bx + c$ and a straight line with equation $y = mx + d$. The curve and the line intersect at points A and B. The vertex of the parabola is at point V. The x-coordinates of the points A, B, and V are given as $x_A = -1$, $x_B = 3$, and $x_V = 1$ respectively. The y-coordinate of the vertex V is $y_V = -4$. Determine the value of a.



a. 1

b. 2

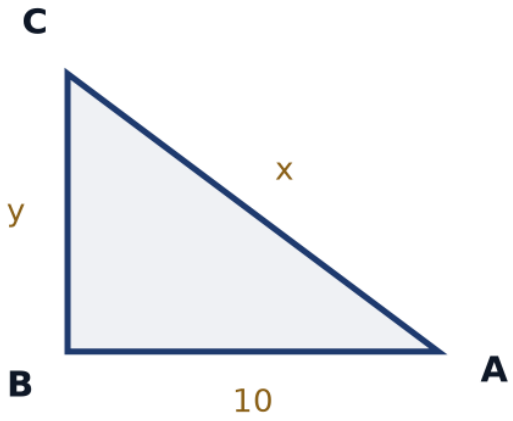
c. -1

d. -2

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34 [Intermediate] • Target: Permutations & Combinations Practice Exercises

In triangle ABC, angle ABC = 90°, AB = 10 cm, and angle BAC = θ. Given that $\sin\theta = \frac{3}{5}$, find the length of AC.



a. 16.7 cm

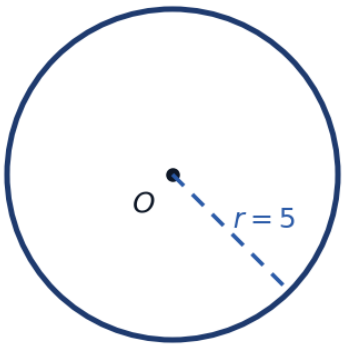
b. 8.0 cm

c. 12.5 cm

d. 6.0 cm

35 [Intermediate] • Target: Permutations & Combinations Practice Exercises

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.



a. 25

b. 30

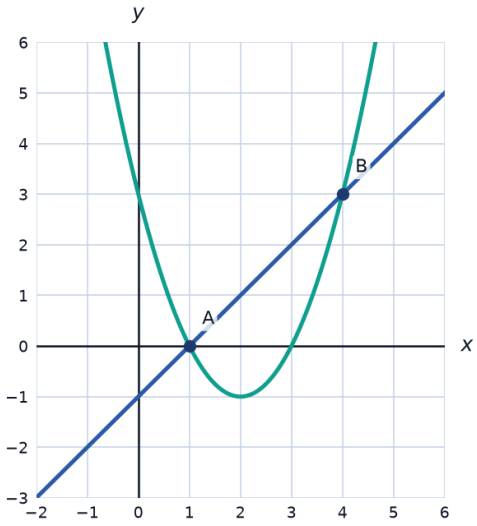
c. 35

d. 40

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36 **[Intermediate]** • **Target: Permutations & Combinations Practice Exercises**

The graph of the quadratic function $y = x^2 - 4x + 3$ is shown. The graph of the linear function $y = x - 1$ is also shown. The graphs intersect at points A and B. Find the coordinates of point A.



a. (1, 0)

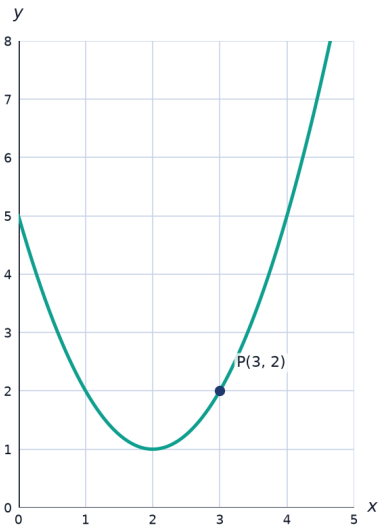
b. (3, 2)

c. (0, 3)

d. (2, 1)

37 **[Intermediate]** • **Target: Permutations & Combinations Practice Exercises**

The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.



a. 2

b. 6

c. -4

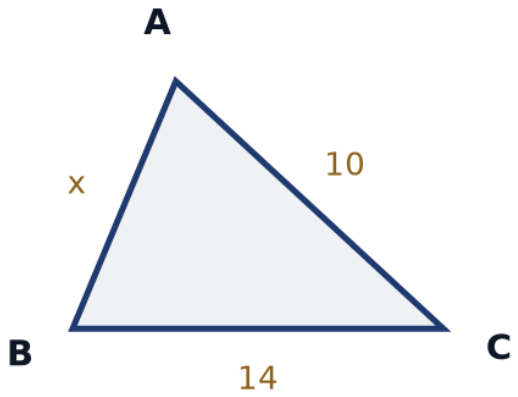
d. $\frac{2}{3}$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities**

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?



a. 10.3

b. 12.1

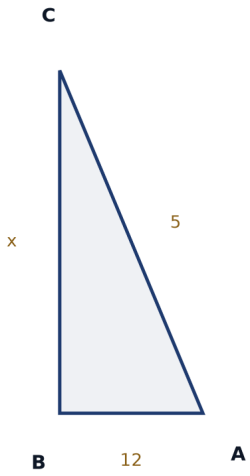
c. 14.8

d. 8.5

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39 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

In the right-angled triangle ABC, $\angle ABC = 90^\circ$. If $AB = 5$ cm and $BC = 12$ cm, find the length of the hypotenuse AC.



a. 10 cm

b. 13 cm

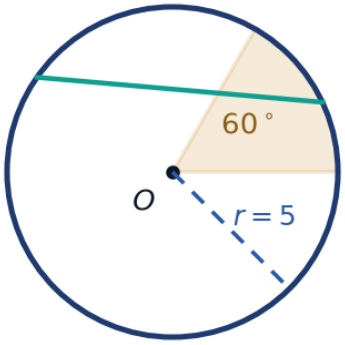
c. 17 cm

d. 144 cm

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40 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?



a. 6 units

b. $6\sqrt{3}$ units

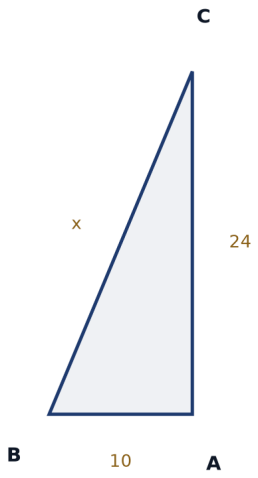
c. 12 units

d. $12\sqrt{3}$ units

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41 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The diagram shows a right-angled triangle ABC, with the right angle at B. The length of side AB is 10 cm and the length of side BC is 24 cm. A semicircle is drawn with AC as its diameter. Calculate the area of the semicircle in terms of π.



a. 169π cm^2

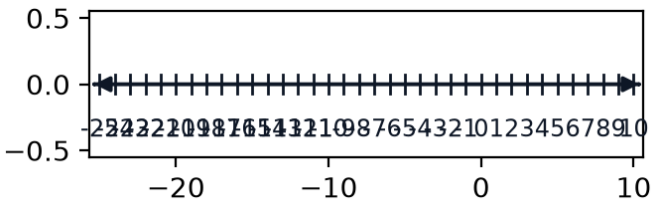
b. 84.5π cm^2

c. 13π cm^2

d. 169 cm^2

42 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

The diagram shows a number line. The distance between the points marked A and B is 15 units. The coordinate of A is -7. Find the possible coordinates of B.



a. -22 and 8

b. 8 and 22

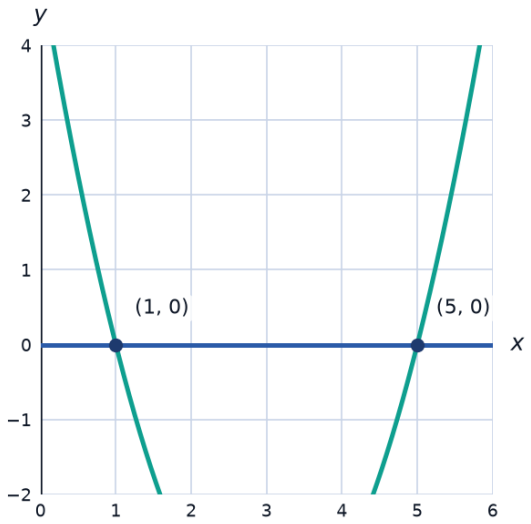
c. -22 and -8

d. 7 and 22

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43 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The diagram shows the graph of the quadratic function $y = x^2 + bx + c$. The graph intersects the x-axis at $(1, 0)$ and $(5, 0)$. Find the values of b and c.



a. $b = -6, c = 5$

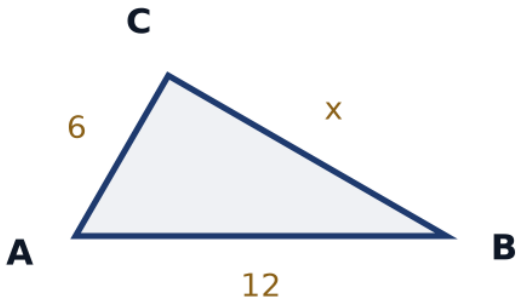
b. $b = 6, c = 5$

c. $b = -6, c = -5$

d. $b = 6, c = -5$

44 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

In a right-angled triangle, one of the acute angles is 30° and the side opposite to this angle is 6 cm. Calculate the length of the hypotenuse.



a. 10 cm

b. 12 cm

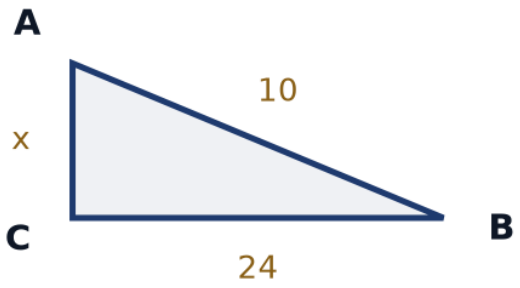
c. 8 cm

d. 6 cm

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45 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.



a. 25

b. 26

c. 27

d. 28

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